In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

FIITJEE **JEE (Main)-2017**

FULL TEST – I

Time Allotted: 3 Hours **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 03 and 10 to 12)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 mark** for wrong answer.

Section-A (04 to 09 and 13 to 30) contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

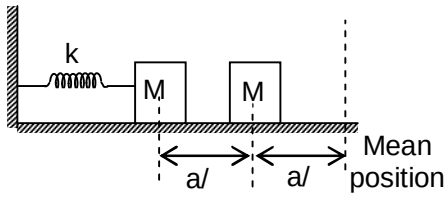
PART – I

SECTION – A

Single Correct Choice Type

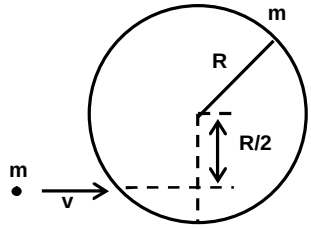
This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. A block of mass M is connected to a spring of force constant k and is placed on a smooth horizontal surface. The block is displaces and compresses the spring by a. The block is left free to move from this position, when the block is at a distance a/2 from the mean position it collides elastically with an identical block. Time the oscillating block takes to reach from extreme to the mean position is



(A) $2\pi\sqrt{\frac{M}{k}}$ (B) $\pi\sqrt{\frac{M}{k}}$ (C) $\frac{5\pi}{6}\sqrt{\frac{M}{k}}$ (D) None

2. A circular loop of mass m and radius R rests flat on a horizontal frictionless surface. A bullet of equal mass moving with a velocity V strikes the hoop and gets embedded in it. The angular velocity of the system after the impact is



(A) $\frac{v}{4R}$ (B) $\frac{v}{6R}$

(C) $\frac{2v}{3R}$ (D) $\frac{3v}{4R}$

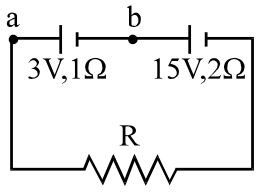
3. A sound source is located somewhere along the x–axis. Experiment shows that the same wave front simultaneously reaches listeners at x = -10 m and x = + 4.0 m. A third listener is positioned along the positive y–axis. What is his y–coordinate (in m) if the same wave front reaches at him at the same instant it does the first two listeners?

(A) $\sqrt{40}$ (B) 3 (C) 7 (D) $\sqrt{30}$

4. Assume that a star has uniform density. Then the gravitational pressure (P) on the surface of the star related to volume (V) of star is.

(A) $P \propto V^{-1/3}$ (B) $P \propto V^{-2/3}$ (C) $P \propto V^{-4/3}$ (D) $P \propto V^{-5/3}$

5. Two batteries one of the emf 3V, internal resistance 1 ohm and the other of emf 15 V, internal resistance 2 ohm are connected in series with a resistance R as shown. If the potential difference between a and b is zero the resistance of R in ohm is

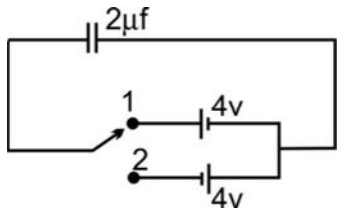


(A) 5 (B) 7

(C) 3 (D) 1

Space for Rough work

6. Heat generated in the circuit when switch is shifted from 1 to 2 is



(A) $32\mu J$ (B) $16\mu J$

(C) $48\mu J$ (D) $64\mu J$

7. A light pointer fixed to one prong of a tuning fork touches a vertical plate. The fork is set vibrating and the plate is allowed to fall freely. Eight complete oscillations are counted when the plate falls through 40 cm. The frequency of tuning fork is (g = 980 cm)

(A) 56 Hz (B) 8 Hz (C) 49 Hz (D) 28 Hz

8. A large tank is filled with water at the rate of 70 cm^3/s. A hole of cross – section area 0.25 cm^2 is punched at the bottom of the tank. The maximum height to which the tank can be filled is

(A) 40 cm (B) 20 cm (C) 10 cm (D) 50 cm

9. Consider the following statements

(i) Young's modulus is numerically equal to the stress which will double the length of a wire

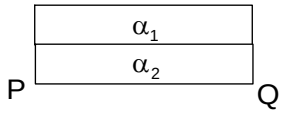
(ii) The surface tension of a liquid decreases due to the presence of insoluble contamination

(iii) Viscosity of gases is greater than that of liquids.

The number of above statements that are true is

(A) zero (B) Three (C) Two (D) One

10. Two composite bar having coefficient of linear expansion α_1 and α_2 respectively, are shown in fig. If the temperature of the bar is raised by t^oC then



(A) Shape of the composite bar is concave toward PQ if $\alpha_1 < \alpha_2$

(B) Shape of the composite bar is convex toward PQ if $\alpha_1 < \alpha_2$

(C) Shape of the bar is concave towards PQ if $\alpha_1 = \alpha_2$

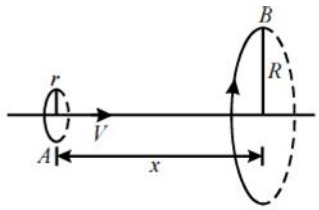
(D) Bar will not bend irrespective of the value of α_1 and α_2

11. The input resistance of a silicon transistor is 100Ω. Base current is changed by 40 μA, which results in a change in collector current by 2 mA. This transistor is used as a common emitter amplifier with a load resistance of $4\ K\Omega$. The voltage gain of the amplifier is

(A) 2000 (B) 3000 (C) 4000 (D) 1000

Space for Rough work

12. Loop A of radius (r << R) moves towards loop B with a constant velocity V in such a way that their planes are always parallel. What is the distance between the two loops (x) when the induced emf in loop A is maximum :



(A) R (B) $\frac{R}{\sqrt{2}}$

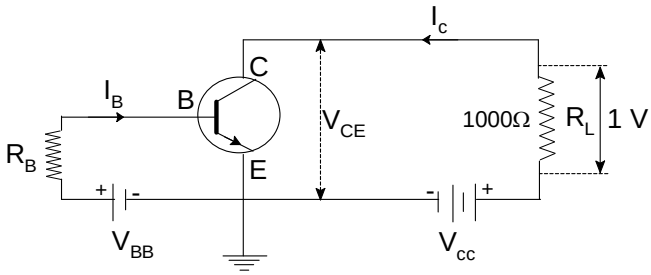
(C) $\frac{R}{2}$ (D) $R\left(1-\frac{1}{\sqrt{2}}\right)$

13. Vernier constant of the Vernier caliper is
(A) least count of Vernier caliper (B) value of 1 division of main scale
(C) value of 1 division of Vernier scale (D) None of these

14. A block of mass *m* is placed on an another rough block of mass *M* and both are moving horizontally with same acceleration *a* due to a force which is applied on the lower block, then work done by lower block on the upper block in moving a distance *s* will be

(A) *Mas* (B) $(m+M)as$ (C) $\frac{M^2}{m}as$ (D) *mas*

15. An N – P – N transistor is connected in common emitter configuration in which collector supply is 9V and the voltage drop across the load resistance of 1000 Ω connected in the collector circuit is 1V. If current amplification factor is (25/26) and the internal resistance of the resistor is 200 Ω.. Then which of the following statement is incorrect.



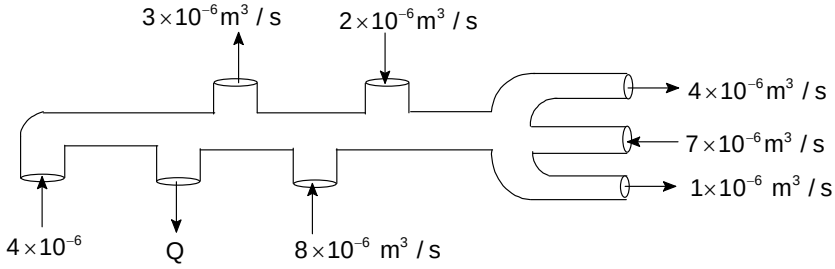
(A) $V_{CE} = 8V$
(B) collector current is 1.0 mA
(C) voltage gain $\frac{50}{23}$, power gain is 4.6
(D) emitter current is 2.04 mA

Space for Rough work

16. The magnifying power of the telescope is found to be 9 and the separation between the lenses is 20 cm for relaxed eye. Then the focal lengths of component lenses are
(A) 18 cm, 2 cm (B) 10 cm, 10 cm
(C) 9 cm, 11 cm (D) None of these

17. In compound microscope
(A) The objective has a shorter focal length
(B) The objective has a shorter aperture
(C) (a) and (b) both are correct
(D) The aperture of objective and eye piece are same

18. The pipe shows the volume flow rate of an ideal liquid at certain time and its direction. What is the value of Q in m^3/s (Assume steady state and equal area of cross section at each opening)



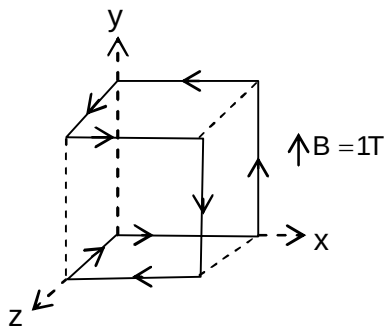
(A) $12\times10^{-6}m^3/s$ (B) $11\times10^{-6}m^3/s$
(C) $9\times10^{-6}m^3/s$ (D) $13\times10^{-6}m^3/s$

19. A parallel beam of monochromatic light is incident on a narrow rectangular slit of width 1 mm. When the diffraction pattern is seen on a screen placed at a distance of 2 m. The width of principal maxima is found to be 2.5 mm. The wave length of light is
(A) 6250 $\overset{o}{A}$ (B) 6200 $\overset{o}{A}$ (C) 5890 $\overset{o}{A}$ (D) 6000 $\overset{o}{A}$

20. In an electromagnetic wave, the amplitude of electric field is 1V/m. The frequency of wave is 5×10^{14}Hz. The wave is propagating along z – axis. The average energy density of electric field in Joule/m^3, will be
(A) 1.1×10^{11} (B) 2.2×10^{-12} (C) 3×10^{-13} (D) 4.4×10^{-14}

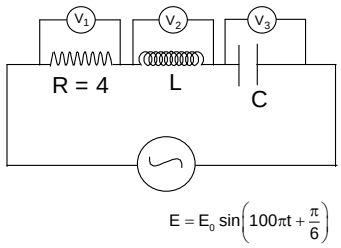
Space for Rough work

21. Figure shows a bent coil with all edges of length 1m and carrying a current 1A, then torque acting on the loop is
(A) zero (B) 3 N-m
(C) 2 N-m (D) 1 N-m



22. Find the minimum thickness of a film which will strongly reflect the light of wavelength 589 nm. The refractive index of the material of the film is 1.25
(A) 118 nm (B) 120 nm (C) 118 mm (D) 120 mm

23. In the figure shown the reading of voltmeters are V_1 = 40 V, V_2 = 40 V and V_3 = 10 V. Find the peak value of emf
(A) $50V$ (B) $50\sqrt{2}V$
(C) $50\sqrt{3}V$ (D) $100\sqrt{2}V$



24. If equation of transverse wave is $y = x_0 \cos 2\pi\left(nt - \frac{x}{\lambda}\right)$. Maximum velocity of particle is twice of wave velocity, if λ is.
(A) $2\pi x_0$ (B) πx (C) $\frac{\pi}{x_0}$ (D) πx_0

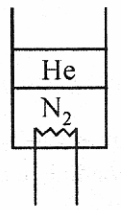
25. The focal length of the objective and the eyepiece of a microscope are 4 mm and 25 mm respectively. If the final image is formed at infinity and the length of the tube is 16 cm. The magnifying power of the microscope will be
(A) -327.5 (B) -32.75 (C) 3.275 (D) 32.75

26. An oscillator has a time period of 3 sec. Its amplitude decreases by 5% each cycle. Its energy decreases in each cycle is
(A) 5% (B) 10% (C) 1.0% (D) 2%

27. A mass 25×10^{-3}Kg is suspended from the lower end of a vertical spring having force constant 25 N/m. The mechanical resistance of the system is 1.5 N.s/m. The mass is displaced vertically and released. Its period of oscillation is
(A) 0.314 (B) 0.628 (C) 3.14 (D) 6.28

Space for Rough work

28. 5 moles of nitrogen gas are enclosed in an adiabatic cylindrical vessel. The piston itself is a rigid light cylindrical container containing 3 moles of Helium gas. There is heater which gives out a power 100 cal/sec to the nitrogen gas. A power of 30 cal/sec is transferred to Helium through the bottom surface of the piston. The rate of increment of temperature of the nitrogen gas assuming that the piston moves slowly:



(A) 2 K/sec (B) 4 K/sec
(C) 6 K/sec (D) 8 K/sec

29. A beam of natural light falls on a system of 6 polaroids, which are arranged in succession such that each polaroid is turned through 30^0 with respect to the preceding one. The percentage of incident intensity that passes through the system will be
(A) 100% (B) 50 %
(C) 30% (D) 12%

30. Which of the following electromagnetic waves have minimum frequency
(A) Micro waves (B) Audible waves
(C) ultrasonic waves (D) Radio waves

Space for Rough work

Chemistry

PART – II

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. The volume of an H_2O_2 solution (volume strength = 11.2) required to convert all the Fe^{+2} present in 4 mole of Fe_3O_4 to Fe^{+3} is:
(A) 2 mL (B) 1 mL (C) 4 L (D) 2 L

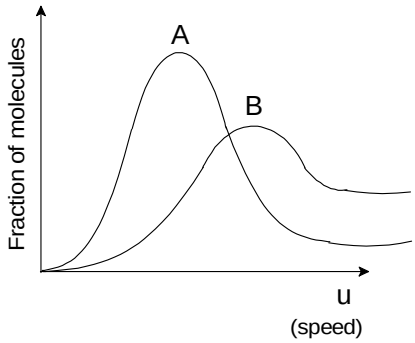
2. In an excited sample of Li^{+2} ions with all the electrons in the same energy state (n_1) produces 3 different spectral lines when all of them are now found to be in 1st excited state. What can't be the orbital angular momentum of the electron present in state n_1?
(A) $\sqrt{12}\hbar$ (B) $\sqrt{2}\hbar$ (C) 0 (D) $\sqrt{20}\hbar$

3. For a reaction $P \longrightarrow Q$, $\frac{-d[P]}{dt}$ increases by a factor of 2 upon increasing the concentration of P by 8 times. The order of the reaction with respect to P is
(A) 3 (B) 1/3 (C) 2 (D) 1/2

4. Assuming that, water vapour is an ideal gas, the internal energy change (ΔU), when 1 mol of water is vaporized at 1 bar pressure & 100^0C, (given: molar enthalpy of vaporization of water at 1 bar & 373 K = 41 $kJmol^{-1}$ & R = 8.314 J mol^{-1} K^{-1}) will be:
(A) 41.00 $kJmol^{-1}$ (B) 3.7904 $kJmol^{-1}$ (C) 4.100 $kJmol^{-1}$ (D) 37.904 $kJmol^{-1}$

Space for Rough work

5. Maxwell's Boltzmann's distribution of molecular speeds for two ideal gases A and B at the same temperature is given below:



Select the incorrect statement:
(A) Fraction of molecules moving with most probable speed of gas A is greater than that of gas B
(B) Most probable speed of molecules of gas B is greater than that of gas A
(C) Rate of diffusion of gas A < gas B under identical conditions
(D) Rate of diffusion of gas B < gas A under identical conditions

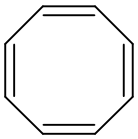
6. In an fcc arrangement of A and B in which A-atoms are at the corners of the unit cell and B – atom at the face centres, one of the A – atom is missing from one corner in unit cells. The simplest formula of the compound is:
(A) A_7B_3 (B) AB_3 (C) A_7B_{24} (D) A_8B_{21}

7. The pH of the following half – cell solution is:
$Pt, H_2(1\,atm)|H^+(HCl); E = 0.25\ V$
(A) 2.21 (B) 8.46 (C) 4.23 (D) 0

8. Which is not correct for physical adsorption?
(A) Adsorption is spontaneous
(B) Both enthalpy and entropy of adsorption are -ve
(C) Adsorption on solid is reversible
(D) Adsorption increases with increase in temperature

Space for Rough work

9. The osmotic pressure of urea solution is 500 mm at 10^0C. The solution is diluted and the temperature raised to 25^0C, when the osmotic pressure is found to be 105.3 mm.
$V_{final} : V_{initial} \simeq X : 1$
X is:
(A) 5 (B) 2 (C) 8 (D) 10

10. Select the correct statement:
(A) Al_2Cl_6 is a planar molecule
(B) I_2Cl_6 is a planar molecule
(C) B_2H_6 is a planar molecule
(D)  is a planar molecule

11. Which of the processes are endothermic?
(i) $Cl(g) + e^- \rightarrow Cl^-(g)$
(ii) $N(g) + e^- \rightarrow N^-(g)$
(iii) $Ne(g) + e^- \rightarrow Ne^-(g)$
(iv) $Ne(g) \rightarrow Ne^+(g) + e^-$
(v) $Li^+(g) \rightarrow Li^+(aq)(\text{hydration})$
(vi) $Cl^-(g) \rightarrow Cl^-(aq)(\text{hydration})$
(A) (ii), (iv) (B) (ii), (iii) (C) (ii), (iii), (iv) (D) (ii), (iii), (iv), (vi)

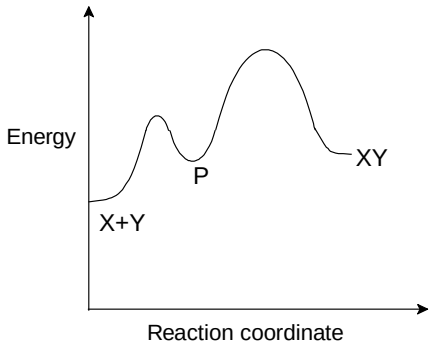
12. The blue colour of dilute solution of alkali metals in ammonia slowly fades because:
(A) electrons get paired
(B) metals form MNH_2 with release of H_2 gas
(C) electrons loose energy eventually because of solvation
(D) electrons get trapped in holes of tetrahedrally arranged NH_3 molecules

Space for Rough work

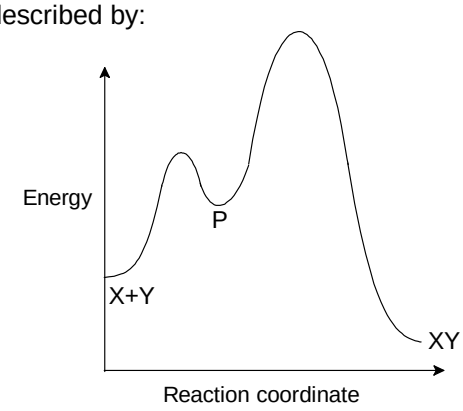
13. Select the correct statement:
(A) D_2O has higher dielectric constant, melting point, boiling point & ionic product than H_2O
(B) D_2O has higher melting point, boiling point & dielectric constant than H_2O but lower value of ionic product
(C) D_2O has higher melting point, boiling point & ionic product but lower dielectric constant than H_2O
(D) D_2O has higher melting point, boiling point but lower dielectric constant & ionic product than H_2O

14. For an exothermic chemical process occurring in two steps as follows:
$(i)\ X + Y \longrightarrow P\,(slow)\ \ (ii)\ P \longrightarrow XY\,(fast)$
The energy diagram for the above process is best described by:

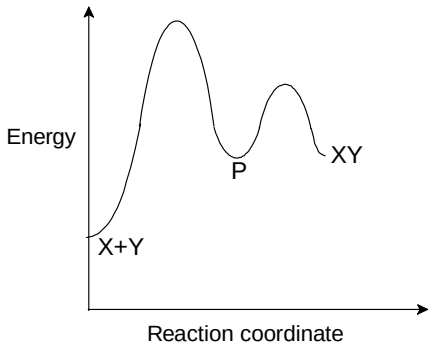
(A)



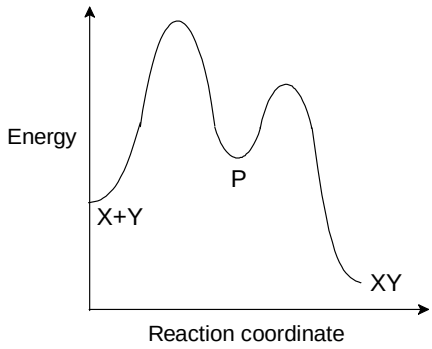
(B)



(C)



(D)



Space for Rough work

15. Which of the following reactions depict the oxidizing behavior of H_2SO_4?

(A) $Ba(NO_3)_2 + H_2SO_4 \longrightarrow BaSO_4 + 2HNO_3$

(B) $2PCl_5 + H_2SO_4 \longrightarrow 2POCl_3 + 2HCl + SO_2Cl_2$

(C) $2NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O$

(D) $2HI + H_2SO_4 \longrightarrow I_2 + SO_2 + 2H_2O$

16. The process in which catalyst is not used:

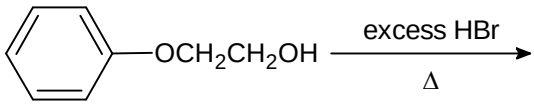
(A) Thermite process
(B) Contact process
(C) Haber's process
(D) Ostwald's process

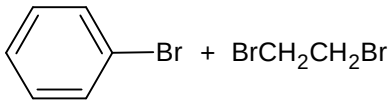
17. Complex compounds which are optically active and doesn't depend upon the orientation of the ligand around metal cation:

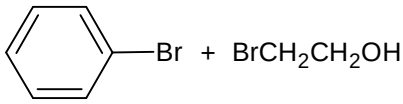
(i) $[CoCl_3(NH_3)_3]$
(ii) $[Co(en)_3]Cl_3$
(iii) $[Co(C_2O_4)_2(NH_3)_2]^-$

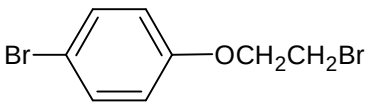
(A) (i), (ii) & (iii)
(B) (ii) & (iii)
(C) only (ii)
(D) all of the above

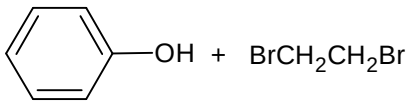
18. What is the product of the following reaction?



(A) 

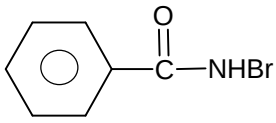
(B) 

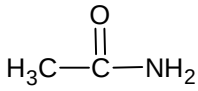
(C) 

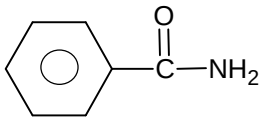
(D) 

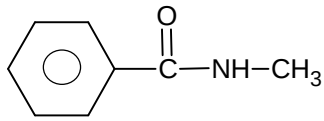
Space for Rough work

19. Which of the following does not give Hoffmann bromamide reaction?

(A) 

(B) 

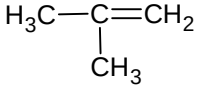
(C) 

(D) 

20. $CH_3 - CH = CH - CH_2 - CO_2H \xrightarrow{\Delta} \underset{(\text{major})}{(X)}$

major product X is:

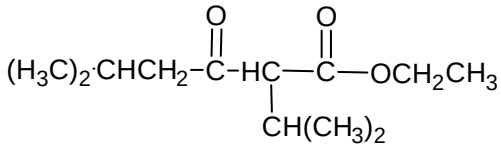
(A) $CH_3 - CH = CH_2$

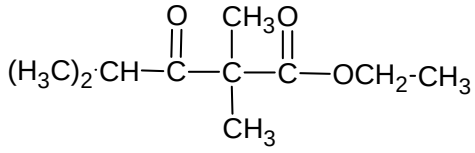
(B) 

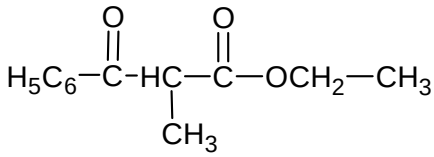
(C) $CH_3 - CH = CH - CH_3$

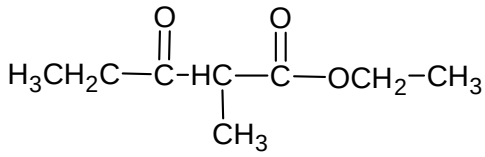
(D) $CH_3 - CH_2 - CH = CH_2$

21. Which of the following keto esters is not likely to have been prepared by a claisen condensation?

(A) 

(B) 

(C) 

(D) 

Space for Rough work

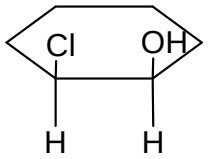
22. $Ph-\overset{\overset{O}{\|}}{C}-OH + CH_3-\overset{18}{O}-H \overset{H^+}{\rightleftharpoons} (y) + H_2O$. y is:

(A) $Ph-\overset{\overset{O}{\|}}{C}-\overset{18}{O}-CH_3$ (Trans esterification)

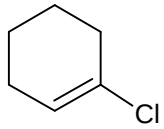
(B) $Ph-\overset{\overset{O}{\|}}{C}-\overset{18}{O}-CH_3$ (Esterification reaction)

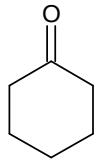
(C) $Ph-\overset{\overset{O}{\|}}{C}-OCH_3$ (hydrolysis)

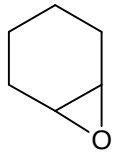
(D) $Ph-\overset{\overset{O}{\|}}{C}-\overset{18}{O}-CH_3$ (saponification)

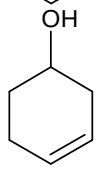
23. 

alc.KOH → (X), product X is:

(A) 

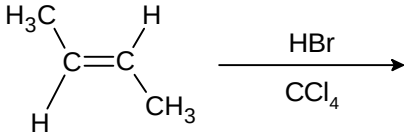
(B) 

(C) 

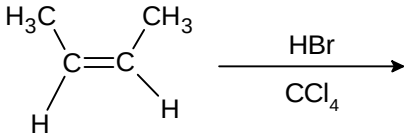
(D) 

Space for Rough work

24. In which of the following reactions, product is racemic mixture?

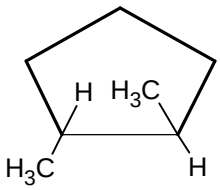
(A) 

(B) $CH_3 - CH_2 - CH = CH_2 \xrightarrow[CCl_4]{HBr}$

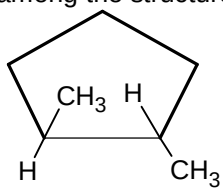
(C) 

(D) All of these

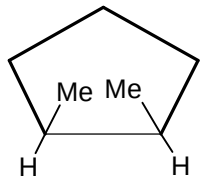
25. Select the enantiomers among the structures given below:



(I)



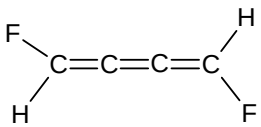
(II)

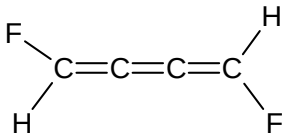


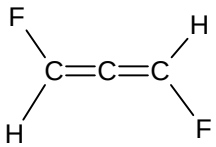
(III)

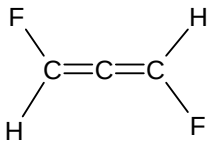
(A) I, II & III (B) I & III (C) I & II (D) II & III

26. Select the correct statement:

(A)  is optically active & non-polar

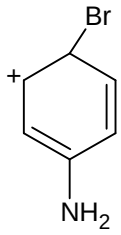
(B)  is optically inactive & polar

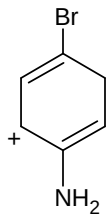
(C)  is optically inactive & non - polar

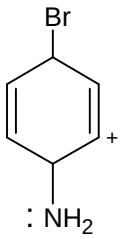
(D)  is optically active & polar

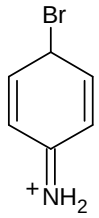
Space for Rough work

27. Which of the following structure is most stable?

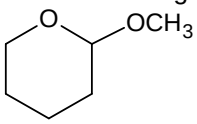
(A) 

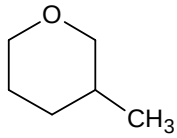
(B) 

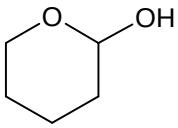
(C) 

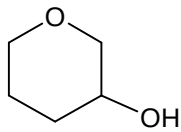
(D) 

28. Which of the following compound give positive Tollen's test?

(A) 

(B) 

(C) 

(D) 

29. A sample of PCl_5 is allowed to decompose in a container against a constant external pressure till following equilibrium achieved:

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

Select the correct option:

(A) The relative vapour density of sample increases till equilibrium is achieved & after that it remains constant

(B) The degree of dissociation $('\alpha')$, is dependent on the total pressure under which the equilibrium is achieved

(C) The degree of dissociation $('\alpha')$ on attainment of the equilibrium is dependent on the total no. of moles of PCl_5 initially taken

(D) The molar mass of the sample increases and the density of the sample remains constant

30. A sparingly soluble salt A_2B_3 has concentration $\left[A^{+3}\right] = SM$ in its saturated aqueous solution. So its solubility (in M) is

$$A_2B_3(s) \rightleftharpoons 2A^{+3}_{(aq)} + 3B^{-2}_{(aq)}$$

(A) S (B) 2S (C) $\frac{S}{2}$ (D) 4S

Space for Rough work

Mathematics

PART – III

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. Solution of differential equation
$\frac{xdx - ydy}{xdy - ydx} = x$, given that y = 0 at x = 1

(A) $y + 1 = \sqrt{x^2 - y^2}$ (B) $y + 1 = \sqrt{y^2 - x^2}$ (C) $y = \sqrt{y^2 - x^2}$ (D) None of these

2. The sum of the coefficient of all the integral power of x in the expansion of $\left(1 + \frac{\sqrt{x}}{2}\right)^{40}$ is

(A) $\frac{3^{40} + 1}{2^{40}}$ (B) $\frac{3^{40} + 1}{2^{41}}$ (C) $\frac{3^{20} + 1}{2^{20}}$ (D) $\frac{3^{20} + 1}{2^{21}}$

3. A family of lies is given by $(2 - 3\lambda)x + (1 + 2\lambda)y - 6 - 12\lambda = 0$, λ being a parameters. The line belonging to this family at maximum distance from the point (4, 3) is L_1 and from the point (-2, -4) is L_2. Then angle between lines L_1 & L_2 is

(A) $\tan^{-1}\left(\frac{11}{23}\right)$ (B) $\tan^{-1}\left(\frac{23}{11}\right)$ (C) $\tan^{-1}\left(\frac{17}{19}\right)$ (D) $\tan^{-1}\left(\frac{19}{17}\right)$

4. If p, q, r are distinct complex number such that $\frac{p}{q} = \frac{q}{r} = \frac{r}{p} = \alpha$, then sum of all values of α will be

(A) 1 (B) -1 (C) 0 (D) None of these

5. If f(x) is differentiable & satisfying

$f(x + y) = f(x) + f(y) + (x^2 + y^2)xy \quad \forall\ x, y \in R$ and $\lim_{x \to 0} \frac{f(x)}{x} = 2$ then f(x) will be

(A) $2 + x^3$ (B) $2x + \frac{x^4}{4}$ (C) $2 + \frac{x^4}{4}$ (D) $2x + x^4 + 2$

Space for Rough work

6. Two numbers x and y are randomly selected on the real number line, such that $0 \le x, y \le 60$. Find the probability that the difference between selected no is not greater than 10.

(A) $\frac{11}{36}$ (B) $\frac{11}{38}$ (C) $\frac{11}{40}$ (D) $\frac{11}{42}$

7. On 8×8 Chess board. If two squares of 1×1 are selected randomly, what is probability that, they have one side common.

(A) $\frac{1}{16}$ (B) $\frac{1}{18}$ (C) $\frac{1}{20}$ (D) $\frac{1}{22}$

8. Number of real value of x satisfy this equation, $2^{2x} + 2^x(3 - 5\cos x) + 1 = 0$

(A) 0 (B) 1 (C) 2 (D) 3

9. $f(x) = \prod_{n=1}^{50}(x-n)^{(51n-n^2)}$ then $\frac{f(51)}{f'(51)}$ equal to

(A) $\frac{1}{1275}$ (B) $\frac{1}{2500}$ (C) $\frac{1}{2550}$ (D) $\frac{1}{5050}$

10. A function defied as $R \to R$, $f(x) = e^{2x^3+3x^2+6x-5}$ will be

(A) One – One & onto (B) One – One & into (C) Many – One & onto (D) Many – One & into

11. $\int \frac{(\cot^2 x - n + 1)}{\cos^n x} = -f(x)[g(x)]^{n-1} + c$ where

$f\left(\frac{\pi}{2}\right) = 1$ then minimum value of $\left[[f(x)]^2 + [g(x)]^2\right]^n$

(A) 1 (B) 2^n (C) 2^{2n} (D) 2^{4n}

12. This equation $\sum_{r=-1}^{m}\left(x - \frac{1}{2^r}\right)^{2n+1} = 0$, has

(A) only one real root (B) has all real root
(C) has 2n real root (D) has (2n +1) real root

Space for Rough work

13. Consider number N = 75600 then sum of divisor of the form (4n + 2), $n \in N$ is
(A) 19840 (B) 29760 (C) 39680 (D) None of these

14. The inverse of the converse of $(p \wedge \sim q) \Rightarrow r$ will be equivalent to
(A) $\sim r \Rightarrow (q \Rightarrow p)$ (B) $\sim r \Rightarrow (\sim p \Rightarrow q)$ (C) $r \Rightarrow (p \Rightarrow q)$ (D) None of these

15. If a, b, c be the sides of a ΔABC and it roots of the equation $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$ are equal then $\operatorname{cosec}^2\frac{A}{2}, \operatorname{cosec}^2\frac{B}{2}, \operatorname{cosec}^2\frac{C}{2}$ are in
(A) AP (B) GP (C) HP (D) None of these

16. Area of the ellipse $3x^2 + 4xy + 3y^2 + 2x + 8y + 6 = 0$
(A) $\frac{\pi}{\sqrt{5}}$ (B) $\sqrt{5}\pi$ (C) $\frac{4\pi}{\sqrt{5}}$ (D) $4\sqrt{5}\pi$

17. $\int_{-\infty}^{0} e^{122x}\left(1-e^{x}\right)^{19} dx$ is equal to
(A) $\frac{19!121!}{71!}$ (B) $\frac{19!121!}{70!}$ (C) $\frac{19!121!}{141!}$ (D) $\frac{19!121!}{140!}$

18. How many 6 letter words can be formed using the letter from the word 'BORINGROAD' if each word has 3 vowels & 3 consonant
(A) 936 (B) 5616 (C) 17280 (D) 33676

19. A man can take a step forward, backward, left or right with equal probability. Then the probability that after nine steps he will be just one step away from his initial position.
(A) $P(E) = 4 \times \left(\frac{15874}{4^9}\right)$ (B) $P(E) = 4 \times \left(\frac{15875}{4^9}\right)$
(C) $P(E) = 4 \times \left(\frac{15876}{4^9}\right)$ (D) $P(E) = 4 \times \left(\frac{15877}{4^9}\right)$

Space for Rough work

20. If $\sum_{r=0}^{n-1}\left[x+\frac{r}{n}\right]=\sum_{r=0}^{n}x$, where n is the positive integer and [x] represent greatest integer of x.
Then total number of possible solution of x is
(A) 1 (B) n -1 (C) n (D) n + 1

21. $\int_0^{\frac{\pi}{8}}\frac{x\,dx}{1+\cos 4x+\sin 4x}=\frac{\pi}{b}\ell n2$ then value of b is
(A) 14 (B) 16 (C) 32 (D) 64

22. $\int_1^e(x+1)e^x\ell nx\,dx=ae^p+be^q+ce^r$ then value of (p + q + r) will be
(A) 2 (B) e + 2 (C) e + 1 (D) None of these

23. If a, b, c represent the sides of triangle ABC and quadratic equation $ax^2+cx+b=0$ does not have two distinct real root then which condition hold's true
(A) $b\geq 4a+2c$ (B) $b\leq 4a-2c$ (C) $b\leq 2c-4a$ (D) $b\geq 2c-4a$

24. $\cos^{-1}\left(x^2-3x+3\right)\geq\left[\pi+2\cos^{-1}\left(x^2\right)\right]$ then total number of possible value of x satisfying this equation is
(A) 0 (B) 1 (C) 2 (D) 3

25. If $A=\begin{bmatrix}1&2&3\\0&1&2\\0&0&1\end{bmatrix}$ & BA = A where B represent 3×3 order matrix. If total number of '1' in matrix A^{-1} & matrix B are p & q respectively then p + q is
(A) 3 (B) 4 (C) 5 (D) 7

26. $\frac{\sin A}{\sin C}=\frac{\sin(C+2B)}{\sin(A+2C)}$ then a sinA, b sinB, csinC are in,
(A) AP (B) GP (C) HP (D) None of these

Space for Rough work

27. The image of the parabola $x^2 = 4y$ in the line x + y = 2 is

(A) $(x-2)^2 = 4(2-y)$ (B) $(y-2)^2 = 4(2-x)$

(C) $(x-2)^2 = 4(2+y)$ (D) $(y-2)^2 = 4(2+x)$

28. If $(\sin 3A + \sin 3B + \sin 3C) + 12(\sin A.\sin B.\sin C) - 3(\sin A + \sin B + \sin C) = 0$ then

(A) $\cos C = \sin A + \sin B$ (B) $\sin C = \cos A + \cos B$

(C) $\sin A = \sin B + \sin C$ (D) None of these

29. $\lim\limits_{x\to 0} \dfrac{\left(e^{2002x}+1\right)-(1+1001x)\left(e^{1001x}+e^{-1001x}\right)}{x\left(e^{x}-1\right)}$

(A) (1001) (B) $(1001)^2$ (C) (2002) (D) $(2002)^2$

30. Equation $2.x^{\left[\left(\log_2^x\right)^2 - \log_2^x + 1\right]} = \sqrt[8]{2}$ has

(A) One real value of x (B) two real value of x

(C) three real value of x (D) None of these

Space for Rough work